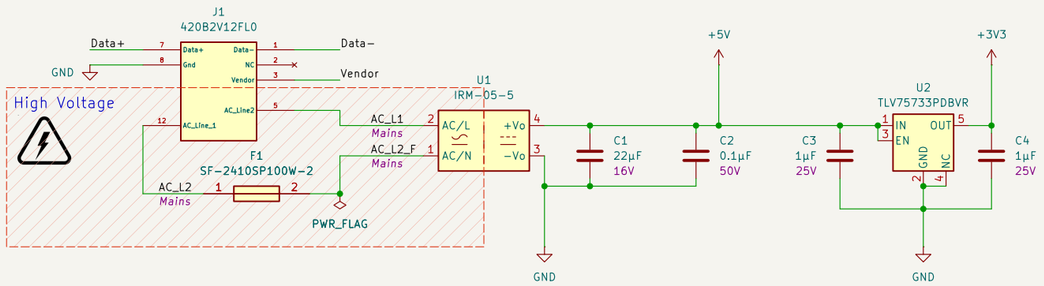
EcoPort Connector and Power Conversion

WARNING: Both line voltage (120V AC Per Line as well as RS-485 5V DC signal is provided through this terminal. Use extreme caution around the high voltage pins and components.

240 AC is converted to 5V - 1A DC through the IRM-05-5 converter. This establishes a 5V rail where needed. 5V DC is converted to 3.3V through the LDO Regulator. This establishes the 3.3V rail used by most components.

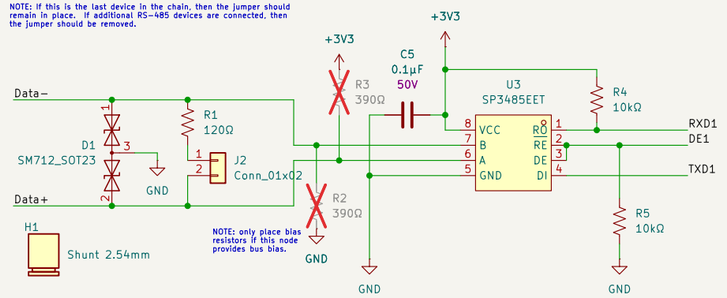
RS485 signal is provided via two Pins 1 and 7 of the connector.



RS-485 Transceiver

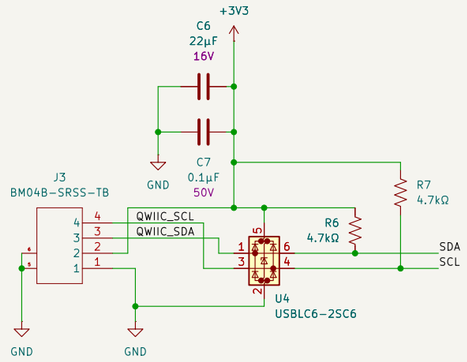
SP3485 converts Data+/- RS-485 signal pair into a UART RX/TX. This is not an auto-sensing transceiver and half-duplex communication must be controlled from the DE/RE pins.

NOTE: If this is the last device in the chain, then the jumper should remain in place. If additional RS-485 devices are connected, then the jumper should be removed.



Stemma QT or Qwiic Connector

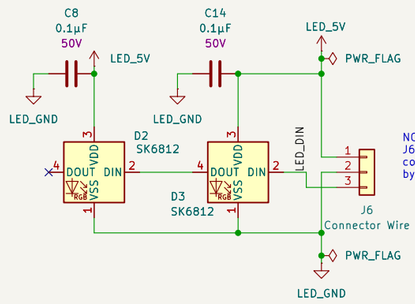
An I2C connector is provided to allow connection to sensor breakout boards. The JST 4-pin connector is compatible with Adafruit Stemma QT or SparkFun Qwiic connectors. Other I2C devices are supported but a connector adaptor may be required. The intention is for a connecting an external sensor, such as a temperature and humidity sensor.



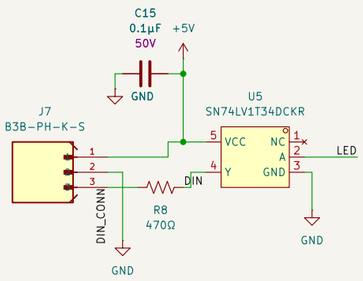
Indicator LEDs

Indicator LEDs using WS2812 integrated, addressable LEDs. Or, in Adafruit paralances, NeoPixels. These are powered by the 5V rail and MCU signal raised from 3.3V to 5V by the logic level shifter.

NOTE: LEDs are placed on a separate board, connected by a 3-wire JST-SH connector from the main board.



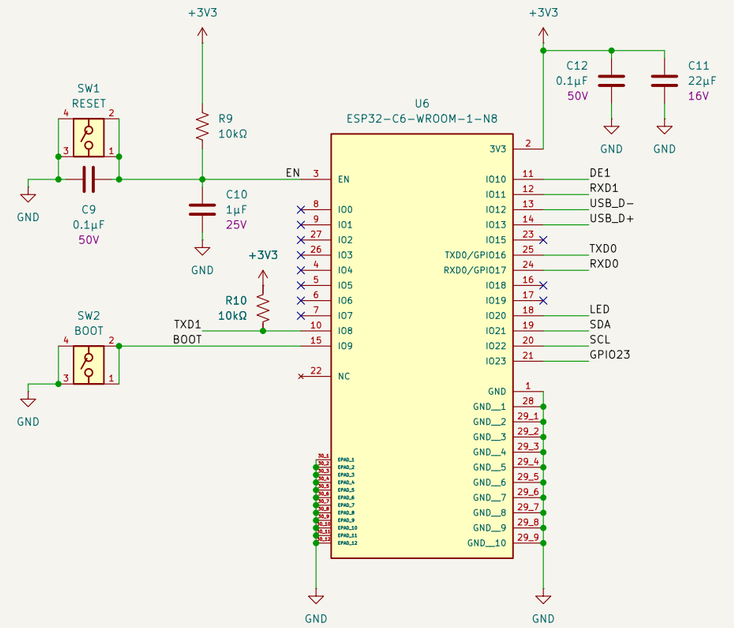
NOTE: J6 and J7 connected by wire



ESP32-C6 Module

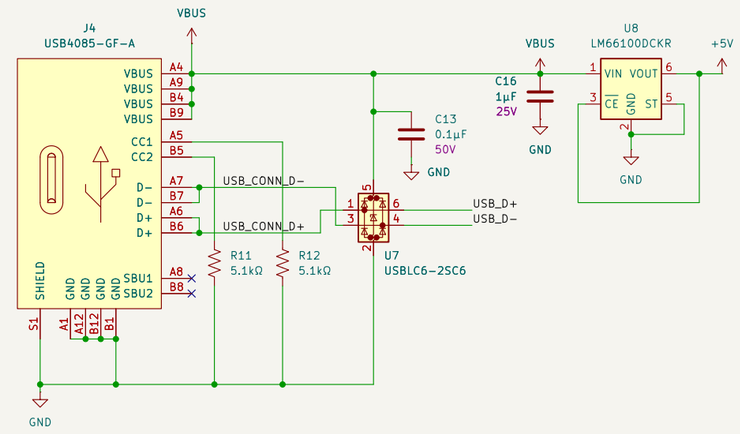
The module is built around the ESP32-C6. It provides 8MB of flash RAM and an on-board PCB antenna.

Two tactile buttons are provided for reset and to control boot options. Tap RESET to reset the Board. Hold BOOT and tap RESET to enter download mode.



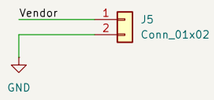
USB-C Connector

USB-C connector is provided for loading firmware or troubleshooting. VBUS feeds +5V through U8, an LM66100 ideal diode, so the board can run from USB alone (bench bring-up). When the IRM-05-5 is powered, U8 blocks reverse current into the host.



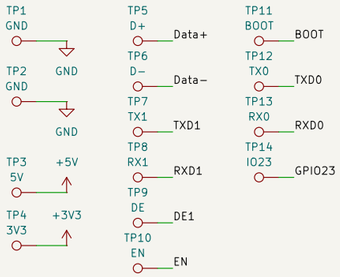
Vendor Data

Unknown vendor data pin output.



Test Points

Test points on nets.



Schematic of the EcoPort Bridge

Jim Rowe

Sheet: /

File: ecoport-bridge-v1.kicad_sch

Title: EcoPort-Bridge UCM

Size: B	Date: 2026-10-12	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/1